



Task 1
Marking Scheme

**Waters of Croatia: From the Emerald
Rivers to the Deep Blue Sea**

EOES2025 Zagreb Croatia
26.04. – 03.05.2025

Problem 1. Ready, Set, Go!

(TOTAL MARKS: 47)

Step 1.1.

*1.1.1. Fill in Table 1.1.1. following the instructions. Count the organisms in Sample A and Sample B, and write the names of the taxonomic groups and the number of individuals in each sample in each taxonomic group in this table. Calculate the benthic density (number of individuals/m²) for each taxonomic group, based on the sample area of 0.049 m². Round the calculation of benthic density to two decimal places. Also fill in the columns where you should check if your organisms are segmented (i.e., if they have distinct body segments along the longitudinal axis) and the column where you should check if the organisms have legs. Indicate this by placing a tick (✓) in the appropriate column for organisms with these characteristics, or a slash (/) if they do not. **(Please note that the number of rows in this table does not indicate the number of individual taxa in each sample.)** (19.25 p)*

Table 1.1.1. Abundance and characteristics of organisms found in Samples A and B.

Sample	Taxonomic group	Number of individuals in the sample	Benthic density (Number of individuals / m ²)	Morphology	
				Segmented	Legs
Sample A	Amphipoda	50 (tolerance: ± 1)	1020.41 (tolerance: 1000 - 1041)	✓	✓
	Ephemeroptera	3 (tolerance: ± 1)	61.22 (tolerance: 41 - 82)	✓	✓
	Diptera	23 (tolerance: ± 1)	469.39 (tolerance: 449 - 490)	✓	/
	Oligochaeta	6 (tolerance: ± 1)	122.45 (tolerance: 102 - 143)	✓	/
	Gastropoda	1 (tolerance: ± 1)	20.41 (tolerance: 0 - 41)	/	/
Sample B	Amphipoda	13 (tolerance: ± 1)	265.31 (tolerance: 245 - 286)	✓	✓
	Plecoptera	4 (tolerance: ± 1)	81.63 (tolerance: 61 - 102)	✓	✓
	Ephemeroptera	10 (tolerance: ± 1)	204.08 (tolerance: 184 - 224)	✓	✓
	Diptera	7 (tolerance: ± 1)	142.86 (tolerance: 122 - 163)	✓	/
	Trichoptera	2 (tolerance: ± 1)	40.82 (tolerance: 20 - 61)	✓	✓
	Coleoptera	2 (tolerance: ± 1)	40.82 (tolerance: 20 - 61)	✓	✓
TOTAL MARKS					19.25 p
each cell in first three columns 0.5 p, last two columns – 0.25 for both correct answers in the same row					

1.1.2. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: D <i>Explanation: In fast-flowing aquatic environments, organisms often need strong physical adaptations to cling to surfaces and resist being swept away by the current. Segmentation (which helps with flexibility and movement) and legs (which allow organisms to crawl, cling, or move quickly) are essential for survival in these habitats. For instance, many macroinvertebrates like mayfly nymphs and caddisfly larvae have specialized structures that allow them to attach to rocks and resist strong flows.</i>	0.5 p

1.1.3. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: B <i>Explanation: A is somewhat misleading: Laterally flattened organisms are typically found in fast-flowing waters near the bottom, where their wide bodies help them cling to surfaces and resist being swept away by the current. B is correct: Streamlined bodies help organisms reduce resistance to the current in fast-flowing waters. C is incorrect: Dorsoventrally flattened organisms can appear in fast-flowing and slow-flowing waters. D is incorrect: In fast-flowing waters, smaller or streamlined organisms are often more efficient at moving through the current.</i>	0.5 p

Step 1.2.

1.2.1. Enter the calculated Simpson Diversity Index (SDI) (rounded to three decimal places) for your two samples (Sample C and Sample D) in the blanks provided and write the letter corresponding to the correct answer in the space provided. (3 p)

	Marks
SDI (Sample C): 0.554	1 p
SDI (Sample D): 0.704	1 p
Sample D (write C or D in the space provided) has greater diversity.	1 p
TOTAL MARKS	3 p

1.2.2. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: B <i>Explanation: A. Incorrect - Improving water quality typically supports a wider variety of species and boosts biodiversity. B. Correct - The introduction of invasive species typically reduces biodiversity by outcompeting or displacing native species, leading to lower diversity. C. Incorrect - Increasing the variety of substrate type in the ecosystem results in increasing habitat complexity (e.g., different substrates, structures, etc.), which promotes species diversity by providing more ecological niches. D. Incorrect – Diverse food sources might support more species.</i>	0.5 p

1.2.3. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: D <i>Explanation: A high Simpson's Diversity Index value (closer to 1) generally reflects an ecosystem where there is a balance between many species, all of which have similar population sizes. This indicates a healthy, diverse ecosystem with a wide variety of species interacting.</i>	0.5 p

Step 1.3.

1.3.1. Enter the calculated proportions of individuals belonging to mayflies (Ephemeroptera), stoneflies (Plecoptera), and caddisflies (Trichoptera) in your two samples (Sample C and Sample D), expressed as percentages (%), rounded to one decimal place, in Table 1.3.1. below. (4 p)

Table 1.3.1. Proportions (%) of mayflies (Ephemeroptera), stoneflies (Plecoptera), and caddisflies (Trichoptera) in Samples C and D.

	Sample C	Sample D
E; Ephemeroptera (%)	0.9 %	23.6 %
P; Plecoptera (%)	0.0 %	12.7 %
T; Trichoptera (%)	0.0 %	7.3 %
EPT; Ephemeroptera + Plecoptera + Trichoptera (%)	0.9 %	43.6 %
TOTAL MARKS		4 p (each cell 0.5 p)

1.3.2. Enter the calculated proportions of individuals belonging to Diptera and Oligochaeta in your two samples (Sample C and Sample D), expressed as percentages (%), rounded to one decimal place, in Table 1.3.2. below. (3 p)

Table 1.3.2. Proportions (%) of Diptera and Oligochaeta in Samples C and D.

	Sample C	Sample D
D; Diptera (%)	20.9 %	10.9 %
O; Oligochaeta (%)	62.3 %	0 %
DO; Diptera + Oligochaeta (%)	83.3 %	10.9 %
TOTAL MARKS		3 p (each cell 0.5 p)

1.3.3. Please enter the letter of the correct answer in the box. (0.5 p)

	Marks
Correct answer: D <i>Explanation: Sample D contains a higher proportion of mayflies, stoneflies, and caddisflies — all of which are sensitive to pollution and indicate better water quality — the higher proportions of Diptera and Oligochaeta in Sample C suggest that this sample might be more tolerant of pollution.</i>	0.5 p

Step 1.4.

1.4.1. Enter your answers in Table 1.4.1. below by placing a tick (✓) in the appropriate column if the organisms have the listed characteristics or a slash (/) if this is not the case. (2.75 p)

Table 1.4.1. Characteristics (feeding types) of organisms found in Samples C and D.

Sample	Taxonomic group	Feeding types	
		Herbivorous/ Detritivorous	Predators
Sample C	Isopoda	✓	/
	Ephemeroptera	✓	✓
	Diptera	✓	✓
	Oligochaeta	✓	/
	Bivalvia	✓	/
	Gastropoda	✓	/ (✓)
Sample D	Amphipoda	✓	✓
	Ephemeroptera	✓	✓
	Plecoptera	✓	✓
	Trichoptera	✓	✓
	Diptera	✓	✓
TOTAL MARKS			2.75 p
0.25 for both correct answers in the same row			

1.4.2. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: B <i>Explanation: Shredders are organisms that break down coarse particulate organic matter, such as decaying plant material. To do this, they need strong mandibles or jaws that allow them to bite and fragment the tough material into smaller pieces. This is a key adaptation for feeding on larger, solid pieces of organic matter. In contrast, organisms with large gills or sucking mouthparts are more specialized for filtering small particles or ingesting liquid food, which is characteristic of other feeding types.</i>	0.5 p

1.4.3. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: D <i>Explanation: A plenty of predators suggest a balanced food web, where predators are feeding on a diverse range of invertebrates. This usually indicates a healthy and stable ecosystem. A. is unlikely, as high levels of algae and detritus are more likely to support herbivores or detritivores, not predators. A healthy predator population typically reflects a balanced ecosystem, not one dominated by algae or detritus. B. is unlikely, because while invasive species can disrupt ecosystems, they typically cause imbalances that reduce native</i>	0.5 p

predator populations. A plenty of predators generally suggests a stable, well-established food web, not one under threat from invasive species. C. is unlikely, as ecosystems with poor water quality and low biodiversity typically have fewer predators. A plenty of predators would indicate a more diverse and stable ecosystem, not one with poor quality and low biodiversity.	
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1.4.4. Please write the letter corresponding to the correct answer in the space provided. (0.5 p)

	Marks
Correct answer: A <i>Explanation: Detritivores and species like oligochaetes that thrive in polluted, low-oxygen conditions typically increase in number when the ecosystem is stressed or degraded. B. is unlikely, as a balanced distribution of feeding types typically suggests a healthy and diverse ecosystem. Pollution usually disrupts this balance, leading to an overrepresentation of certain types, such as detritivores, rather than a diverse range of feeding types. C. is unlikely as scrapers and predators, like stoneflies, are typically found in clean, well-oxygenated water. A high proportion of these organisms would usually indicate a healthy ecosystem, not a polluted one. D. is unlikely, because while filter-feeders and shredders can be found in various ecosystems, they are generally present in healthy environments with abundant organic material. A high proportion of these organisms alone wouldn't necessarily indicate pollution.</i>	0.5 p

Step 1.5.

1.5.1. Enter the results of your calculations (rounded to two decimal places) for the volume of water passing through each drift sampler during the sampling period, the drift density (number of organisms/m³) for each taxonomic group in each drift sample, and the averages for all parameters in Table 1.5.1. below. (8 p)

Table 1.5.1. Results of drift calculations based on the data obtained during and after the collection of the drift samples.

	Volume of water filtered through the net	Ephemeroptera drift density (Number of organisms/m ³)	Amphipoda drift density (Number of organisms/m ³)	Diptera drift density (Number of organisms/m ³)
Drift net 1	5.97 m ³	≈ 3.69 individuals/m ³	≈ 6.04 individuals/m ³	≈ 2.01 individuals/m ³
Drift net 2	8.73 m ³	≈ 3.21 individuals/m ³	≈ 4.36 individuals/m ³	≈ 1.15 individuals/m ³
Drift net 3	5.30 m ³	≈ 2.83 individuals/m ³	≈ 4.15 individuals/m ³	≈ 3.21 individuals/m ³
AVERAGE	6.67 m ³	3.24 individuals/m ³	4.85 individuals/m ³	2.12 individuals/m ³
TOTAL MARKS				8 p (each cell 0.5 p)

Calculation:

1. Volume of water filtered by DRIFT NET 1:

- Calculate the volume of water passing through the drift sampler during the sampling period.
- **Time (in seconds):** 45 minutes = 2700 seconds
- **Flow rate (m³/s):**
Flow rate = Area × Flow velocity = 0.00442 m² × 0.5 m/s = 0.00221 m³/s
- **Total volume filtered:**
Total volume = Flow rate × Time = 0.00221 m³/s × 2700 s = 5.967 m³

Drift Density Calculation:

- For each taxonomic group, calculate the drift density (individuals per m³):
 - **Ephemeroptera:**
Drift Density = 22 individuals/5.967 m³ ≈ 3.69 individuals/m³
 - **Amphipoda:**
Drift Density = 36 individuals/5.967 m³ ≈ 6.04 individuals/m³
 - **Diptera:**
Drift Density = 12 individuals/5.967 m³ ≈ 2.01 individuals/m³

2. Volume of water filtered by DRIFT NET 2:

- Calculate the volume of water passing through the drift sampler during the sampling period.
- **Time (in seconds):** 47 minutes = 2820 seconds
- **Flow rate (m³/s):**
Flow rate = Area × Flow velocity = 0.00442 m² × 0.7 m/s = 0.003094 m³/s
Total volume filtered:
Total volume = Flow rate × Time = 0.003094 m³/s × 2820 s = 8.725 m³
Drift Density Calculation:
- For each taxonomic group, calculate the drift density (individuals per m³):
 - **Ephemeroptera:**
Drift Density = 28 individuals/8.725 m³ ≈ 3.21 individuals/m³
 - **Amphipoda:**
Drift Density = 38 individuals/8.725 m³ ≈ 4.36 individuals/m³
 - **Diptera:**
Drift Density = 10 individuals/8.725 m³ ≈ 1.15 individuals/m³

3. Volume of water filtered by DRIFT NET 3:

- Calculate the volume of water passing through the drift sampler during the sampling period.
- **Time (in seconds):** 50 minutes = 3000 seconds
- **Flow rate (m³/s):**
Flow rate = Area × Flow velocity = 0.00442 m² × 0.4 m/s = 0.001768 m³/s
Total volume filtered:
Total volume = Flow rate × Time = 0.001768 m³/s × 3000 s = 5.304 m³
Drift Density Calculation:
- For each taxonomic group, calculate the drift density (individuals per m³):
 - **Ephemeroptera:**
Drift Density = 15 individuals/5.304 m³ ≈ 2.83 individuals/m³
 - **Amphipoda:**
Drift Density = 22 individuals/5.304 m³ ≈ 4.15 individuals/m³
 - **Diptera:**
Drift Density = 17 individuals/5.304 m³ ≈ 3.21 individuals/m³

AVERAGE: Total volume filtered: 6,665 m³, Ephemeroptera drift density ≈ 3.24 individuals/m³, Amphipoda drift density ≈ 4.85 individuals/m³, Diptera drift density ≈ 2.12 individuals/m³

1.5.2. Enter the results of your calculations (rounded to three decimal places) for the drift propensity for each taxonomic group (Ephemeroptera, Amphipoda, Diptera) in the Table 1.5.2. below. (1.5 p)

Table 1.5.2. Results of drift propensity calculations based on benthic and drift density data

	Benthic density (Number of individuals/m ²)	Drift density (Number of individuals/m ³)	Drift propensity
Ephemeroptera	429	10	0.023
Amphipoda	3531	24	0.007
Diptera	122	12	0.098
TOTAL MARKS			1.5 p (each cell 0.5 p)

1.5.3. Based on your results, draw conclusions about which groups are more likely to drift downstream. Enter your answers in Table 1.5.3. **Use the symbols >, <, or = to indicate the relationships between the taxonomic groups in terms of their drift propensity (1.5 p):**

Table 1.5.3. Relationships between the taxonomic groups in terms of their drift propensity.

	Marks
Ephemeroptera <u><</u> Diptera Amphipoda <u><</u> Ephemeroptera Diptera <u>></u> Amphipoda	1.5 p (each correct answer 0.5 p)

Problem 2. Salts from Croatian Saltworks

(TOTAL MARKS: 37)

Step 2.1. Chemical reactions for determination of iodine species

2.1.1. Fill the Table 2.1.1. (Answer Sheet) with your results by marking the observed change with an X. (3.5 p)

Table 2.1.1. Observations

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T1	X						0.5 p
T2	X						0.5 p
T3	X						0.5 p
T4	a)		X				0.5 p
	b)					X	0.5 p
T5	a)			X			0.5 p
	b)				X		0.5 p
TOTAL MARKS							3.5 p

2.1.2. Write the balanced chemical reactions in test tubes T4a and T5a (write states of matter) into Table 2.1.2. (Answer Sheet). (1 p)

Table 2.1.2. Balanced chemical reactions in T4 and T5

Test tube	Balanced chemical reactions	Marks
T4 a)	$\text{KI (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{AgI (s)} + \text{KNO}_3 \text{ (aq)}$	0.5 p
T5 a)	$\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{AgCl (s)} + \text{NaNO}_3 \text{ (aq)}$	0.5 p
TOTAL MARKS		1 p

Step 2.2. Testing salt solutions

2.2.1. Fill in the Table 2.2.1. on the Answer Sheet by marking the observed change with an X. (1.5 p)

Table 2.2.1. Observed changes in sample Salt 1

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T6						X	0.5 p
T7	X						0.5 p
T8						X	0.5 p
TOTAL MARKS							1.5 p

2.2.2. Fill in the Table 2.2.2. on the Answer Sheet by marking the observed change with an X. (1.5 p)

Table 2.2.2. Observed changes in sample Salt 2

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T9						X	0.5 p
T10						X	0.5 p
T11						X	0.5 p
TOTAL MARKS							1.5 p

2.2.3. Fill in the Table 2.2.3. on the Answer Sheet by marking the observed change with an X. (1.5 p)

Table 2.2.3. Observed changes in sample Salt 3

Test tube	dark blue solution	brown solution	light yellow precipitate	white precipitate	precipitate dissolved	no change	Marks
T12						X	0.5 p
T13	X						0.5 p
T14						X	0.5 p
TOTAL MARKS							1.5 p

2.2.4. In Table 2.2.4. mark with an X the correct form of iodine that is present in table salt samples or mark no iodine if it is not present. (1.5 p)

Table 2.2.4. Form of iodine present in the samples Salt 1, Salt 2 and Salt 3

Salt sample	I ₂	I ⁻	IO ₃ ⁻	No iodine	Marks
Salt 1			X		0.5 p
Salt 2				X	0.5 p
Salt 3			X		0.5 p
TOTAL MARKS					1.5 p

Confirmation of results. Signature of lab assistant: _____

Step 2.3. Measuring iodine in table salts

2.3.1. Determine the oxidation states of iodine in iodine molecule, iodate ion and iodide ion. Write the answer in the Table 2.3.1. on the Answer Sheet. (1.5 p)

Table 2.3.1. Oxidation states of iodine in iodine molecule (I_2), iodate ion (IO_3^-) and iodide ion (I^-)

	Oxidation state of iodine	Marks
Iodine (I_2)	0	0.5 p
Iodate (IO_3^-)	V	0.5 p
Iodide (I^-)	-I	0.5 p
TOTAL MARKS		1.5 p

2.3.2. In the reaction of iodine and thiosulfate ion, determine what species is being reduced and what species is being oxidized. Indicate your answer by X in appropriate cell in Table 2.3.2. on the Answer Sheet. (1 p)

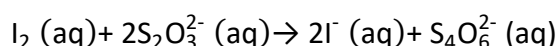


Table 2.3.2. Species being reduced and species being oxidized in the reaction of iodine and thiosulfate ion

	I_2	$S_2O_3^{2-}$	Marks
Species being reduced	X		0.5 p
Species being oxidized		X	0.5 p
TOTAL MARKS			1 p

2.3.3. What is the role of starch solution in the titration of iodine with the standard solution of sodium thiosulfate? Enter the corresponding letter in Answer Sheet. (0.5 p)

- a) starch is used as indicator in the detection of endpoint of reaction involving iodine solution.
- b) starch is used to stabilize the solution of iodine.
- c) starch is used to achieve acid medium needed for the reduction of iodate ions into iodine.

Table 2.3.3. Please write the letter corresponding to the correct answer in the space provided.

	Marks
Correct answer: <u> a) </u>	0.5 p

2.3.4. In the Table 2.3.4. mark with an X which salt samples have you chosen based on the previous results. (1 p)

Table 2.3.4. Samples chosen for quantitative analysis

Salt sample	Samples chosen for analysis	Marks
Salt 1	X	0.5 p
Salt 2		
Salt 3	X	0.5 p
TOTAL MARKS		1 p

Confirmation of results. Signature of lab assistant: _____

2.3.5. In the Table 2.3.5. write which procedure you have chosen (A, B, C or none), and raise the red card so the lab assistant can check your answer. Only after that can you start with the experiment. The lab assistant will check answers for tasks 2.2.4, 2.3.4. and 2.3.5. and sign them if the answers are correct. If the answers are incorrect, you will not get points for these tasks, but the lab assistant will provide you with correct answers. (1.5 p)

Table 2.3.5. Sample(s) and procedure(s) for quantitative analysis of salt samples

	Procedure	Marks
Salt 1	B	0.5 p
Salt 2	NONE	0.5 p
Salt 3	B	0.5 p
TOTAL MARKS		1.5 p

Confirmation of results. Signature of lab assistant: _____

Correct answers

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Salt 1	Iodine species: _____ ;	procedure _____
Salt 2	Iodine species: _____ ;	procedure _____
Salt 3	Iodine species: _____ ;	procedure _____
		Signature of lab assistant: _____

(Total marks for 2.3.6 + 2.3.7 + 2.3.8: 12 p)

2.3.6. If you analyzed sample Salt 1, in Table 2.3.6. on the Answer sheet write the volumes of $\text{Na}_2\text{S}_2\text{O}_3$ used for titration, calculate the mean volume and circle the volumes used in your calculation.

If you didn't analyze sample Salt 1 write "/" in the cells of Table 2.3.6.

Write down the concentration of $\text{Na}_2\text{S}_2\text{O}_3$ as indicated on the bottle _____

Table 2.3.6. Volumes of $\text{Na}_2\text{S}_2\text{O}_3$ used for titration of the salt sample

Titration number	V(Na ₂ S ₂ O ₃) (mL)	Mean value V̄(Na ₂ S ₂ O ₃) (mL)	Marks
		8.30	
TOTAL MARKS			6 p

Point distribution:

- 1 point if the mean volume is correctly calculated
- Precision: 1 point if chosen volumes do not differ by more than 0.20 mL
- Accuracy: 4 points if average value of $\bar{V}(\text{Na}_2\text{S}_2\text{O}_3)$ does not differ from the true value by more than 0.20 mL, 3 points if it differs between 0.20 mL and 0.30 mL, 2 points for difference between 0.30 mL and 0.40 mL, 1 point for difference between 0.40 mL and 0.50 mL, zero points if they differ from true value by 0.50 mL or more.
- If only 2 titrations were made the student will be penalized with 2 points, If only 1 titration is made the student will be penalized with 4 points
- Minimum amount of points for this question is 0
- If 3 titrations are performed, but 2 volumes are used for calculation of mean volume, student is not penalized

2.3.7. . If you analyzed sample Salt **2**, in Table 2.3.7. on the Answer sheet write the volumes of $\text{Na}_2\text{S}_2\text{O}_3$ used for titration, calculate the mean volume and circle the volumes used in your calculation.

If you didn't analyze sample Salt **2** write "/" in the cells of Table 2.3.7.

Write down the concentration of $\text{Na}_2\text{S}_2\text{O}_3$ as indicated on the bottle _____

Table 2.3.7. Volumes of $\text{Na}_2\text{S}_2\text{O}_3$ used for titration of the salt sample

Titration number	$V(\text{Na}_2\text{S}_2\text{O}_3)$ (mL)	Mean value $\bar{V}(\text{Na}_2\text{S}_2\text{O}_3)$ (mL)	Marks
TOTAL MARKS			

Salt sample 2 does not contain iodine in any form, so this should be left empty.

2.3.8. If you analyzed sample Salt **3**, in Table 2.3.8. on the Answer sheet write the volumes of $\text{Na}_2\text{S}_2\text{O}_3$ used for for titration, , calculate the mean volume and circle the volumes used in your calculation.

If you didn't analyze sample Salt **3** write "/" in the cells of Table 2.3.8.

Write down the concentration of $\text{Na}_2\text{S}_2\text{O}_3$ as indicated on the bottle _____

Table 2.3.8. Volumes of $\text{Na}_2\text{S}_2\text{O}_3$ used for titration of the salt sample

Titration number	V(Na ₂ S ₂ O ₃) (mL)	Mean value V̄(Na ₂ S ₂ O ₃) (mL)	Marks
		4.22	
TOTAL MARKS			6 p

Point distribution:

- 1 point if the mean volume is correctly calculated

- Precision: 1 point if chosen volumes do not differ by more than 0.20 mL
- Accuracy: 4 points if average value of $\bar{V}(\text{Na}_2\text{S}_2\text{O}_3)$ does not differ from the true value by more than 0.20 mL, 3 points if it differs between 0.20 mL and 0.30 mL, 2 points for difference between 0.30 mL and 0.40 mL, 1 point for difference between 0.40 mL and 0.50 mL, zero points if they differ from true value by 0.50 mL or more.
- If only 2 titrations were made the student will be penalized with 2 points, If only 1 titration is made the student will be penalized with 4 points
- Minimum amount of points for this question is 0
- If 3 titrations are performed, but 2 volumes are used for calculation of mean volume, student is not penalized

2.3.9. Calculate the molar mass (in g mol^{-1} units) of sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$), iodine molecule (I_2), potassium iodate (KIO_3) and potassium iodide (KI). Write the answer in the Table 2.3.9 (Answer sheet). (2 p)

Calculation

Table 2.3.9. Molar mass of sodium thiosulfate, iodine molecule, potassium iodate and potassium iodide

	$M / \text{g mol}^{-1}$	Marks
$\text{Na}_2\text{S}_2\text{O}_3$	158.11	0.5 p
I_2	253.81	0.5 p
KIO_3	214.00	0.5 p
KI	166.00	0.5 p
TOTAL MARKS		2 p

2.3.10. In the table 2.3.10. on the Answer sheet calculate the amount (in moles) of sodium thiosulfate used for titration of samples. Indicate the sample you are performing calculation for (sample Salt 1, Salt 2 or Salt 3). (1 p)

Table 2.3.10. Calculation of moles of sodium thiosulfate used for titration of samples

	Marks
Sample: <u> 1 </u> $n(\text{Na}_2\text{S}_2\text{O}_3) = c(\text{Na}_2\text{S}_2\text{O}_3) \times \bar{V}(\text{Na}_2\text{S}_2\text{O}_3) = 8.30 \times 10^{-6} \text{ mol (or } 8.30 \times 10^{-3} \text{ mmol)}$	0.5 p
Sample: <u> 3 </u> $n(\text{Na}_2\text{S}_2\text{O}_3) = c(\text{Na}_2\text{S}_2\text{O}_3) \times \bar{V}(\text{Na}_2\text{S}_2\text{O}_3) = 4.22 \times 10^{-6} \text{ mol (or } 4.22 \times 10^{-3} \text{ mmol)}$	0.5 p
Sample: _____	
TOTAL MARKS	1 p

2.3.11. In the table 2.3.11. on the Answer sheet calculate the amount (in moles) of iodine species (I^- , I_2 or IO_3^- , according to the form of iodine which is present in the given sample) present in the initial sample solution (250 mL). Indicate the sample you are performing calculation for (sample Salt 1, Salt 2 or Salt 3). (2 p)

Table 2.3.11. Calculation of moles of iodine species present in the sample(s)

	Marks
Sample: <u> 1 </u> $n(IO_3^-, 50 \text{ mL}) = \frac{1}{6} n(S_2O_3^{2-}) = 1.38 \times 10^{-6} \text{ mol}$ $n(IO_3^-, 250 \text{ mL}) = \frac{250}{50} n(IO_3^-) = 6.92 \times 10^{-6} \text{ mol} = n(KIO_3, 250 \text{ mL})$	1 p
Sample: <u> 3 </u> $n(IO_3^-, 50 \text{ mL}) = \frac{1}{6} n(S_2O_3^{2-}) = 7.03 \times 10^{-7} \text{ mol}$ $n(IO_3^-, 250 \text{ mL}) = \frac{250}{50} n(IO_3^-) = 3.52 \times 10^{-6} \text{ mol} = n(KIO_3, 250 \text{ mL})$	1 p
Sample: _____	

TOTAL MARKS	2 p
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2.3.12. . In the table 2.3.12. on the Answer sheet calculate the mass of iodine species (KI, I₂ or KIO₃, according to the form of iodine which is present in the given sample) in the initial sample solution (250 mL). Indicate the sample you are performing calculation for (sample Salt 1, Salt 2 or Salt 3). (1 p)

Table 2.3.12. Calculation of the mass of iodine species present in the sample(s)

	Marks
Sample: __1__ $m(\text{KIO}_3) = M(\text{KIO}_3) \times n(\text{KIO}_3, 250 \text{ mL}) = 1.48 \text{ mg}$	0.5 p
Sample: __3__ $m(\text{KIO}_3) = M(\text{KIO}_3) \times n(\text{KIO}_3, 250 \text{ mL}) = 0.75 \text{ mg}$	0.5 p
Sample: _____	
TOTAL MARKS	1 p

2.3.13. In the table 2.3.13. on the Answer sheet calculate the fraction (as mg per kg of salt) of iodine species (KI, I₂ or KIO₃, according to the form of iodine which is present in the given sample) in the sample(s), rounded to 1 decimal places. Indicate the sample you are performing calculation for (sample Salt 1, Salt 2 or Salt 3). (1 p)

Table 2.3.13. Calculation of the amount of iodine species in the sample(s) as mg of iodine species per kg of salt

	Marks
Sample: __1__ $m(\text{KIO}_3) = \frac{m(\text{KIO}_3)}{m(\text{salt sample})} = 29.6 \text{ mg/kg}$	0.5 p
Sample: __3__ $m(\text{KIO}_3) = \frac{m(\text{KIO}_3)}{m(\text{salt sample})} = 15.0 \text{ mg/kg}$	0.5 p
Sample: _____	
TOTAL MARKS	1 p

2.3.14. Based on the results, mark on the map the saltworks from which the salt samples (Salt 1, Salt 2 and Salt 3) came, if you know that the saltwork in Ston does not add iodine into their salt, and that saltwork in Pag adds more iodine into their salt than the saltwork in Nin. (2 p)

	Marks
The map shows the outline of Croatia. Four locations are marked with dots and labeled in boxes: Zagreb (inland, north), Pag (coastal, west), Nin (coastal, west of Pag), and Ston (coastal, south-east). Each label box contains the name and a number in a red box: Pag 1, Nin 3, and Ston 2. Zagreb has no number.	2 p

If 2 salt samples are correctly assigned to corresponding saltworks : 1 point

If 3 salt samples are correctly assigned to corresponding saltworks : 2 point

Problem 3. Bending Light in Aqueous NaCl Solution

(TOTAL MARKS: 36)

3.1.1. In Figure 3.2 a) and b) the incoming beam (gray dashed line) reaches the boundary between two media (full black line), in this case between sapphire and ethanol. (Two protractors are also drawn on the Figure, one on each side of a particular medium.) Based on the data from Table 3.1 and Snell's law, draw the refracted rays for both cases on the Answer Sheet. (The accuracy of the drawing may not be worse than ± 1 degree.) Show your calculation. (2 p + 2 p)

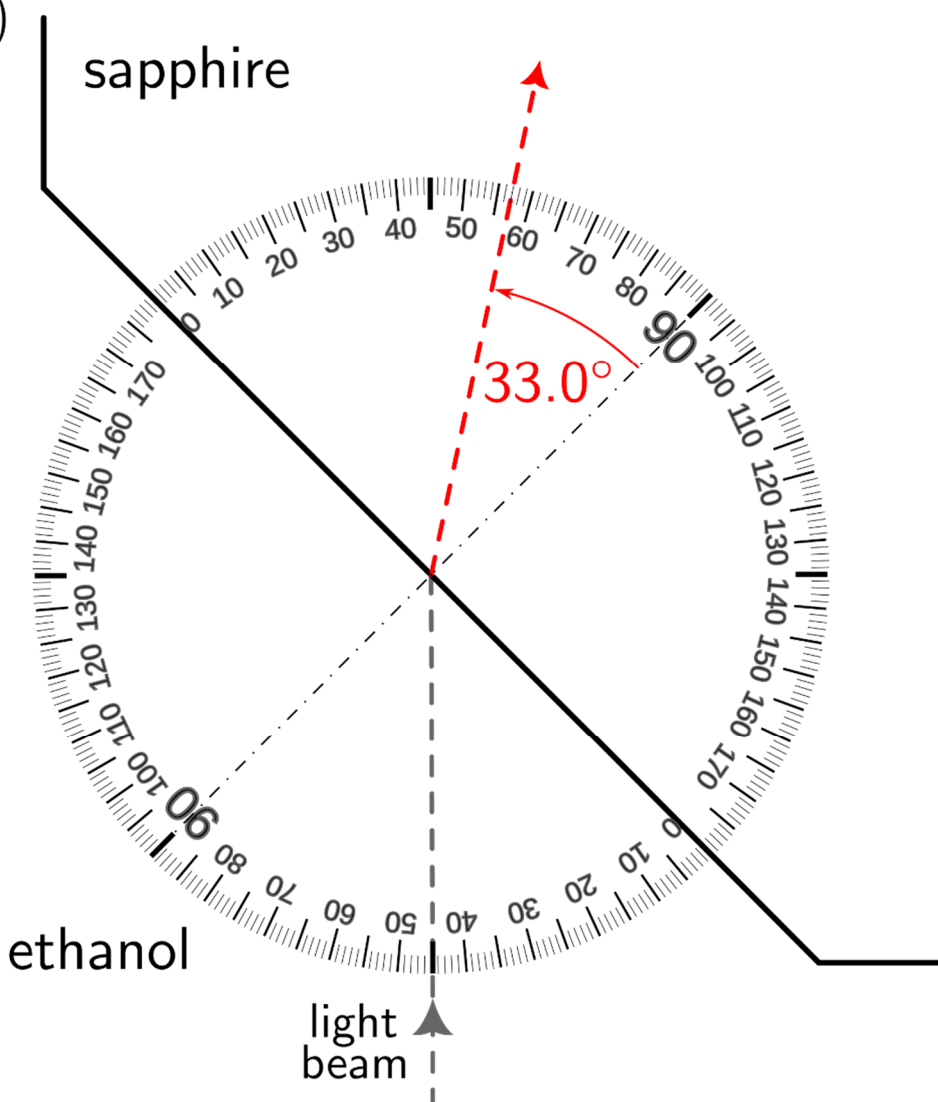
a)  sapphire ethanol light beam 33.0°	Marks 2 p
Calculation $\alpha_1 = 45^\circ$ $(1.362) \sin 45^\circ = (1.770) \sin \alpha_2$ $\alpha_2 = 32.96^\circ \cong 33^\circ$ 0.5 point for correct incident angle value 0.8 point for correct refracted angle value 0.7 point for correctly drawn line	

Figure 3.2. Full black line: the boundary between ethanol and sapphire. Gray dashed line: the incoming light beam (the arrow indicates the direction of the propagation).

a) The beam reaches the boundary from ethanol.

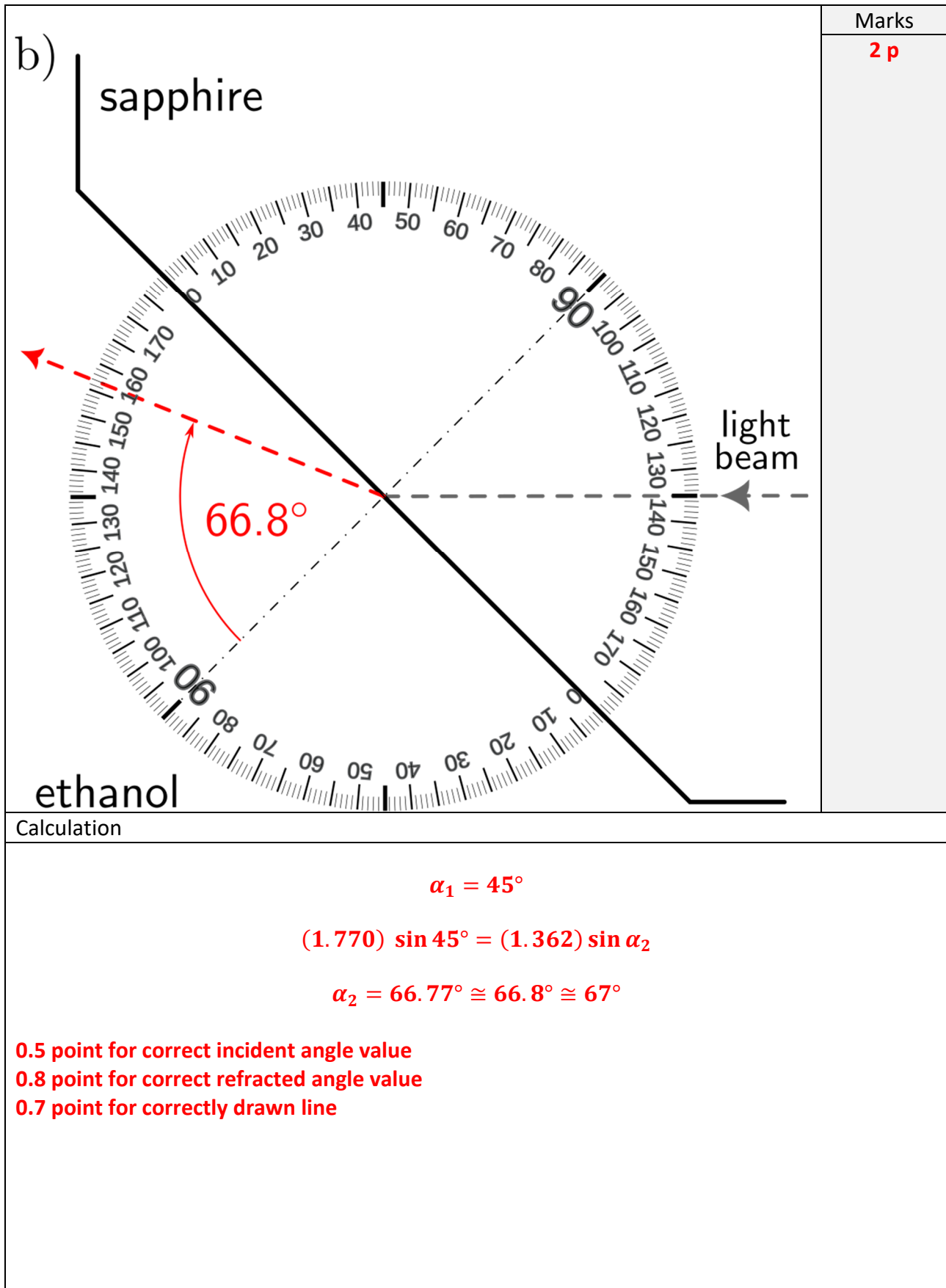
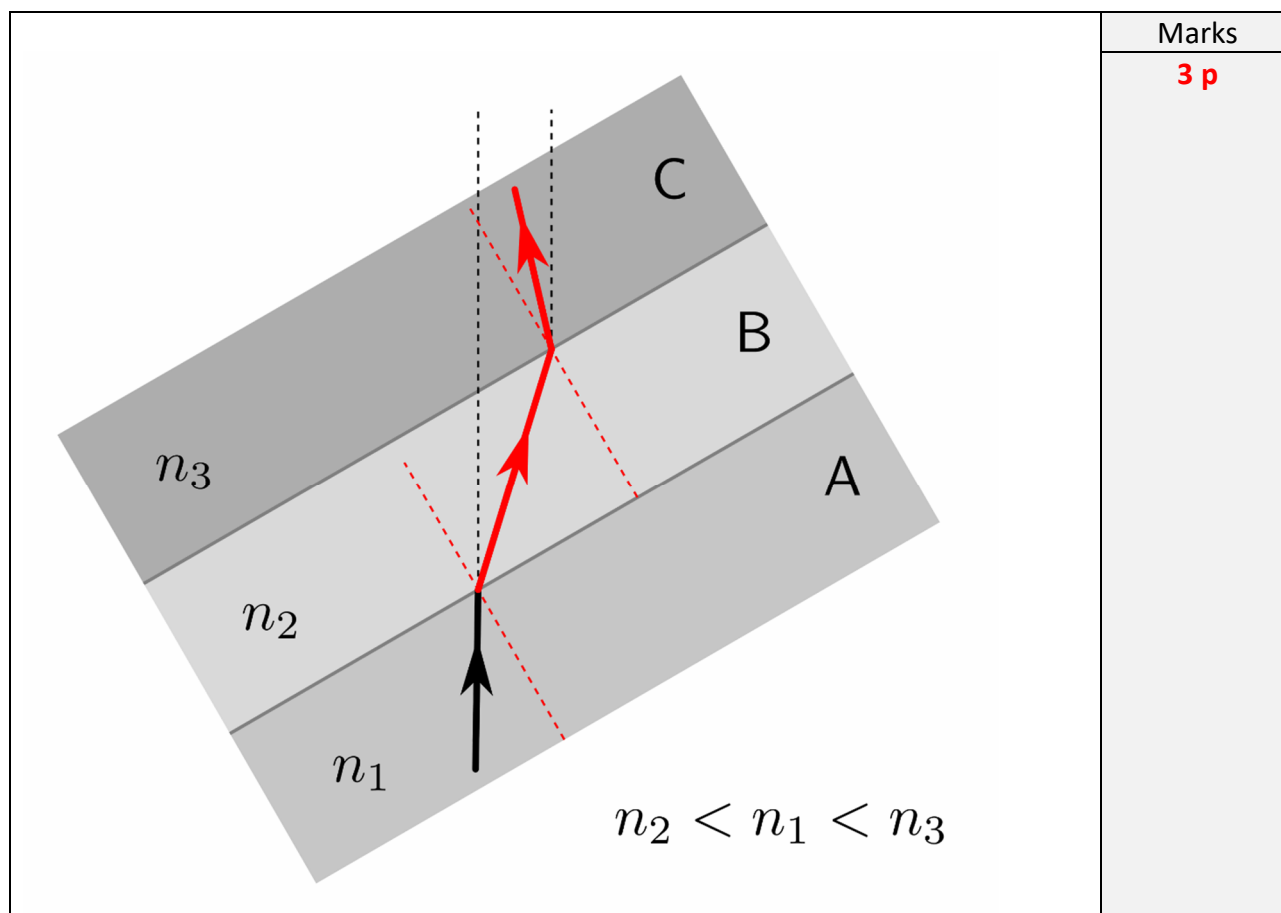


Figure 3.2. Full black line: the boundary between ethanol and sapphire. Gray dashed

line: the incoming light beam (the arrow indicates the direction of the propagation).

b) The beam reaches the boundary from sapphire.

3.1.2. In Figure 3.3, a ray of light is successively refracted at several boundaries of different media. For the incoming ray in medium A, as indicated in the figure, qualitatively sketch the approximate path of the light ray in media B and C. Pay special attention to the relative direction of the rays in media A and C. (5 p)



1p if A-to-B is correct

1p if B-to-C is correct

1p if C is closer to normal than A

Figure 3.3. The black line represents the direction of the incoming light ray in medium A (with refractive index n_1).

3.2.1. (1 p), 3.2.2. (3 p), 3.2.3. (9 p) and 3.2.5. (9 p)

Table 3.3. The table for entering the measured and the calculated values.

$\alpha(\text{deg})$	x exact (cm)	y exact (cm)	$x(\text{cm})$	$y(\text{cm})$	$d(\text{cm})$ 0%	$d(\text{cm})$ 20%
0						
5						
10						
15						
20						
25						
30						
35						
40						
45						
50						
55						
60						
65						
70						
75						
1p for first two columns (x exact, y exact) 3p for second two columns (x, y) 9p points for “d (cm) 0%” column 9p points for last column “d (cm) 20%”						Marks
						22 p

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(Note: if it is not possible to perform all the measurements, the last row of the table may be left empty.)

3.2.4. Add the appropriate amount of table salt to the tub so that you get a 20 % mass fraction aqueous solution. (The '20 % mass fraction' means: for 24 g of demineralized water add 6 g of salt.) Write the mass and height of demineralized water, and mass of added salt. (1 p)

Write the mass of demineralized water and added salt in the Answer Sheet.	Marks
Mass of water: between 1 kg and 1.5 kg	0.2 p
Height of water: it should agree with mass of water above	0.1 p
Mass of added salt: it should be 25% of mass of water above	0.7 p
TOTAL MARKS	1 p

3.3.1. Using a linear fit, determine the slope of $Y = Y(X)$, and hence the refractive indices for demineralized water and the aqueous table salt solution. Write the obtained values (together with their errors with two significant digits). (2 p)

		Marks
Demineralized water	$n = \underline{\underline{1.2759}} \pm \underline{\underline{0.0061}}$	2 p
20% aqueous NaCl solution	$n = \underline{\underline{1.3237}} \pm \underline{\underline{0.0054}}$	2 p
TOTAL MARKS		4 p

Subtract 1 p for every incorrect value i.e. error

Accepted range: $1.22 < n < 1.40$

Accepted error(s): less than 1%

Index of refraction of the solution must be larger than that of demineralized water

Differences between indices of refraction must be less than 5%

3.3.2. In the experiment and the calculation, we ignored the influence of the plexiglass tub walls. Based on the value of the refractive index of plexiglass from Table 3.1, estimate whether the true values of the refractive indices of the liquids are **higher**, **the same**, or **lower** than the ones measured in the experiment. Check the answer (☒) in the table below. (2 p)

			Marks
True values of the refractive indices of the liquids are	☐ larger	than the measured values of the refractive indices of the liquids	2 p
	☐ the same		
	☒ smaller		